

CSEM Research Proposal:

**Simulation of Multiphase Flow and Transport
in Partially Melted Materials***

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Proposal Abstract

Volcanism at the Earth’s surface is a geologically important process that originates due to partial melting of the Earth’s convecting mantle and subsequent melt segregation. Partial melting generates the oceanic and continental crust and determines the large-scale chemical evolution of the Earth. Moreover, similar processes may occur on other terrestrial planets. Most of the Earth’s mantle is generally below its melting point, while partial melting occurs in the following three tectonic settings.

Mid-ocean ridges: Upwelling of the convecting mantle beneath mid-ocean ridges leads to decompression melting and generates the oceanic crust.

Hot-spots: Thermal anomalies induce melting beneath hot-spots and generate ocean islands and oceanic plateaus as well as flood basalt provinces on the continents.

Subduction zones: The addition of fluids that lower the melting point induces flux-melting above the dehydrating down-going oceanic crust and generates the continental crust.

This project will lead to the development of a new code for simulation of partial melting processes that can be applied to these scenarios. The proposal is divided into three main components. We will first develop and mathematically analyze the mathematical framework for computational simulation of evolving mantle flow. Our mathematical model consists of two parts, a Darcy-Stokes system describing the mechanical flow subproblem and an advection-diffusion system for the transport subproblem. The Darcy-Stokes system was previously studied and modeled by Arbogast, Hesse and Taicher in [3], and it is based on mixture theory. The solid mantle is modeled as a slowly creeping viscous flow governed by the compressible Stokes equation subject to compaction. As material melts, it flows according to Darcy’s law. Transitions between non porous single phase flow and two-phase porous flow will occur dynamically in time. In order to deal with partial melting and possible no melting degeneration, we use a scaled formulation that allows for regions of no melt. Within the flow field, the two components of mass and the thermodynamic energy of the system are governed by transport equations. Average concentration of one component and total internal energy per unit volume are used as transport variables. Given these two transport variables, a flash calculation is performed to equilibrate the thermodynamic phase behavior of the system. Binary eutectic phase behavior will be implemented in the process of the flash calculation. For this transport system, we propose a new formulation of the transport equations by introducing two partition function coefficients which are determined in terms of the flash calculation. They will be used to simplify the coupling between the mass and energy transport variables.

In the second part of the proposal, we discuss spatial and time discretization for our coupled system. A conforming mesh of quadrilaterals will be used in our computations. Quadrilateral meshes require fewer elements than traditional triangular meshes, and yet can still follow mildly complex geometry. For ease of implementation, we will use a logically rectangular mesh which maintains natural ordering. We discretize the Stokes equations using a new version of the Bernardi-Raugel spaces and a variant of the lowest order reduced $H(\text{div})$ -approximation AC space [6] for the Darcy part of flow. Both of these elements are defined directly on quadrilaterals without mapping from a reference element. Second order accuracy is achieved in the velocity field with only a piecewise constant pressure approximation. The convergence and stability of the new Bernardi-Raugel element will also be studied in this project. In addition, the mixed finite element discretization will be modified for local mass conservation. For the transport part, implicit WENO-AO methods [5, 4] will be used to approximate the transport of temperature and chemical components to third order accuracy. Energy and mass transport will be solved as a coupled system. The third order implicit

Radau-IIA Runge-Kutta method will be used to advance time, and the corresponding nonlinear system of equations will be solved by Newton iteration at each time step.

Our third and final component of the proposal is the development of a efficient and parallel simulation code. Parallel communication and linear solvers will be implemented using PETSc [7]. Two different algorithms will be used to solve our system sequentially or iteratively coupled. An additional phase code will be integrated to handle the phase behavior, modified from the matlab code developed by Hesse [14]. The code will be able to perform large scale simulation of partially molten mantle dynamics.

We summarize our research contributions in the following paragraphs, organized by CSEM Research Area.

Area A: Applicable mathematics. A new and simpler local mass conservative mixed finite element scheme is under development for the Darcy-Stokes system describing partially melted mantle flow. For the Stokes part of this system, we are also developing a modified Bernardi-Raugel like element defined using the lowest order direct serendipity space. The stability and convergence of this new element will be studied, and will be extended to three dimensions in later application.

Partition functions will be introduced as an alternative to Katz’s approach [16] of decoupling the transport system. The transport will be solved using two different algorithms, to be studied and compared, one computing sequentially fixed in time values for flow velocities and then transported components, while the other iterates between flow and transport within a monolithic quasi-Newton solver. Even though our formulation achieves local mass conservation for the flow and transport parts separately, we will analyze it in the context of the complete coupled system, for both solution algorithms. We will also analyze the error for the flow and transport parts separately, as well as the propagation of errors within the complete discrete coupled system.

Area B: Scientific computing. A PETSc implementation of the proposed algorithms will be completed. Efficient parallel implementation of implicit WENO-AO and mixed finite element methods will be researched, so the code will handle large-scale, parallel simulations. The suitability of the linearized formulation of the partition functions will be studied from an algorithmic point of view, to see if it indeed both converges and improves the overall performance of the code. A proper stabilizer for the transport problem will be investigated. We will begin by considering the suitability of a Lax-Friedrichs type numerical flux, at least in the applications we are targeting. The two different solution algorithms, sequential and iterative coupling, will also be compared and evaluated for their efficiency and accuracy.

Our code will employ the implicit WENO-AO method, which to our knowledge will be its first use in full-scale mantle simulation. We expect that it will control and reduce numerical oscillation in the solution better than is seen for competing codes.

Area C: Application to Earth and planetary science. The partition function formulation will be studied to see if it indeed encapsulates enough of the physics that sequential coupling could be useful in the applications. Our new partition function approach will be compared with the standard approach proposed by Katz [16]. The introduction of eutectic phase behavior will provide a better approximation of small amounts of partial melting in the Earth’s mantle.

We expect to see an improved approximation of the dynamics of partial melting in simulations of melt extraction beneath mid-ocean ridges and its geochemical consequences. In particular, we aim to simulate a viscous corner flow with decompression melting and a conductive lid that channels the melt towards the mid-ocean ridge. We will study the effect of chemical heterogeneities with reduced melting point on melt generation and extraction, and their resultant geochemical signatures in the erupted lavas.

1 Introduction

Volcanism at the Earth’s surface is a geologically important process that originates due to partial melting of the Earth’s convecting mantle and subsequent melt segregation. Partial melting generates the oceanic and continental crust and determines the large-scale chemical evolution of the Earth. Moreover, similar processes may occur on other terrestrial planets. Most of the Earth’s mantle is generally below its melting point, while partial melting occurs in the following three tectonic settings.

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Several different projects have developed codes/packages for mantle simulation. The ASPECT project [8] of the Computational Infrastructure for Geodynamics (CIG) community is developing a popular package for 2D and 3D problems. It uses a relatively straightforward formulation of the governing equations, which does not strictly allow for regions of zero melt. ASPECT is developed in a modular manner, which enables the feasibility of freely defining finite element shape functions and mesh adaptivity using P4EST. Standard Taylor-Hood elements were implemented [11] for the Stokes part of the equations, and various finite elements are used for the rest of the equations. Another notable mantle simulation project produced RHEA [10], which focuses on petascale parallel simulation of the entire Earth, but does not simulate partial melting effects. RHEA is built on ALPS, which enables trilinear hexahedral finite element and high-order spectral elements with discontinuous Galerkin discretization.

This project will have three main components. We will first develop and mathematically analyze the mathematical framework for computational simulation of evolving mantle flow. We use a scaled formulation that allows for regions of no melt, and we propose a new formulation of the transport equations using partition function coefficients. Second, we discretize the equations using a new version of the Bernardi-Raugel spaces for Stokes flow and a variant of the lowest order reduced $H(\text{div})$ -approximation AC space [6] for the Darcy part of flow. Implicit WENO-AO methods to accurately approximate the transport of temperature and chemical components. In addition, Binary eutectic phase behavior will be implemented. These features distinguish this project from other projects, since for the first time to our knowledge, they will be used in a major simulator. Our third and final component is the development of this efficient and parallel simulation code. Parallel communication and linear solvers will be implemented using PETSc. It will implement the flow and transport algorithms, as well as a method for handling the phase behavior [14] for a large scale simulation of partially molten mantle dynamics fitting in above scenarios.

2 Derivation of the Governing Equations

In this section, we summarize the standard model governing mantle dynamics and ice flows [15, 19, 16] over a domain Ω . It is based on mixture theory, and it has three main components. First is a set of equations describing the mechanics of flow of the system. It consists of coupled Darcy and Stokes equations. The second main component is the conservative transport of mass and energy.

It consists of a set of advection-diffusion equations. Finally, there is a set of conditions describing phase equilibrium, accomplished through a *flash calculation*.

We assume that the system has two components (labeled 1 and 2) which can appear within three phases. Two of the phases are solid (labeled 1 and 2), and the other is fluid (a liquid, labeled f). We further assume that the solid phases are pure, consisting only of the corresponding component. However, both solid phases can melt into the fluid phase and the fluid phase can solidify. While in the fluid phase, the two components can react with each other.

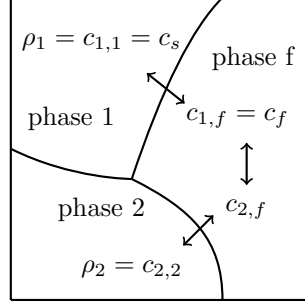


Figure 1: An elementary representative volume (REV).

We show in Figure 1 an elementary representative volume (REV) of our system. Within the REV, $M_{i,j}$ and $V_{i,j}$ are the mass and volume of component $i \in \{1, 2\}$ in phase $j \in \{1, 2, f\}$. Let $c_{i,j}$ denote the concentration of component i in phase j . Within phase 1,

$$c_s = c_{1,1} = \rho_1 = \frac{M_{1,1}}{V_1}, \quad c_{2,1} = 0 = M_{2,1}, \quad (2.1)$$

and within phase 2,

$$c_{1,2} = 0 = M_{1,2}, \quad c_{2,2} = \rho_2 = \frac{M_{2,2}}{V_2}, \quad (2.2)$$

where ρ_i is the density of component i . For the fluid phase, the concentration of component i is

$$c_f = c_{1,f} = \frac{M_{1,f}}{V_f}, \quad c_{2,f} = \frac{M_{2,f}}{V_f}. \quad (2.3)$$

The phase porosities of the system are the relative volume fractions. They are defined by

$$\phi_f = \frac{V_f}{V}, \quad \phi_1 = \frac{V_1}{V}, \quad \phi_2 = \frac{V_2}{V}, \quad (2.4)$$

where $V = V_1 + V_2 + V_f$ is the total volume of the REV. Sometimes it is useful to refer to the total solid porosity

$$\phi_s = \phi_1 + \phi_2 = 1 - \phi_f. \quad (2.5)$$

Given a physical quantity ψ that may exist in each phase, in mixture theory, its bulk phase volume average is defined to be

$$\bar{\psi} = \phi_f \psi_f + \phi_1 \psi_1 + \phi_2 \psi_2. \quad (2.6)$$

We fix the notation for some other quantities of interest in this problem, distributed in phase $j \in \{1, 2, f\}$. The pressure is denoted $p_s = p_1 = p_2$ in the solid phases and p_f in the fluid. In general, $\rho_j = \rho_j(p_j)$; however, we will assume that the densities ρ_1 and ρ_2 are constant. We assume that phase j flows under the velocity $\mathbf{v}_j$, and, of course, $\mathbf{v}_s = \mathbf{v}_1 = \mathbf{v}_2$. We work in a regime where $\phi_f |\mathbf{v}_f|$ and $\phi_s |\mathbf{v}_s|$ are of similar magnitude. Finally, let the phase specific internal energies (i.e., internal energies per unit mass) be denoted e_j .

2.1 Mechanics of flow as a Darcy-Stokes system

The mixture model describing the mechanics of flow is based on that given by McKenzie [19]. The solid mantle is modeled as a slowly creeping viscous flow governed by the compressible Stokes equation subject to compaction. As material melts, it flows according to Darcy's law. The overall system is based on mixture theory, and it transitions dynamically in time from a non porous single phase Stokes flow to a two-phase porous medium [3]. The governing equations [19, 3] are written in terms of the the fluid and solid pressures p_f and p_s , and the relative velocity $\mathbf{v}_r = \mathbf{v}_f - \mathbf{v}_s$ as

$$\phi_f \mathbf{v}_r + \frac{k_0 \phi_f^{2+2\Theta}}{\mu_f} \nabla(p_f - \rho_f \mathbf{g}) = 0, \quad (2.7)$$

$$\nabla \cdot (\phi_f \mathbf{v}_r + \mathbf{v}_s) = 0, \quad (2.8)$$

$$\nabla \bar{p} - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = \bar{\rho} \mathbf{g}, \quad (2.9)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_f(p_f - p_s) = 0, \quad (2.10)$$

where the deviatoric stress of the mixture is

$$\hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = 2\mu_s \phi_s (\mathcal{D}\mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I}) \quad (2.11)$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient. Moreover, the relative permeability of the Darcy flow is $k_0 \phi_f^{2+2\Theta}$ for some spatially varying coefficient k_0 and parameter $\Theta \in [0, 1/2]$, the fluid and solid viscosities are μ_f and μ_s , and $\mathbf{g}$ is the gravitational acceleration constant vector.

The Darcy part will mathematically degenerate in regions where the porosity is zero. To handle this mathematical issue, we use the formulation developed by Arbogast, Hesse, and Taicher [3]. We first define the pressure potentials as

$$q_f = p_f - \rho_f g z \quad \text{and} \quad q_s = p_s - \rho_f g z, \quad (2.12)$$

where $g = |\mathbf{g}|$ and z is the depth, so $\mathbf{g} = g \nabla z$. Note that ρ_f is used to define both potentials, so that with $q = \bar{q}$,

$$p_f - p_s = q_f - q_s = \frac{1}{1 - \phi_f} (q_f - q).$$

We can then rewrite the system (2.7)–(2.10) as

$$\phi_f \mathbf{v}_r + \frac{k_0 \phi_f^{2+2\Theta}}{\mu_f} \nabla q_f = 0, \quad (2.13)$$

$$\mu_s \nabla \cdot (\phi_f \mathbf{v}_r) + \frac{\phi_f}{1 - \phi_f} (q_f - q) = 0, \quad (2.14)$$

$$\nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1 - \phi_f) \rho_r \mathbf{g}, \quad (2.15)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f}{1 - \phi_f} (q_f - q) = 0. \quad (2.16)$$

Scaled variables are defined,

$$\tilde{\mathbf{v}}_r = \phi_f^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_f = \phi_f^{1/2} q_f, \quad (2.17)$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla(\phi_f^{-1/2} \tilde{q}_f) = 0, \quad (2.18)$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0, \quad (2.19)$$

$$\nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1 - \phi_f) \rho_r \mathbf{g}, \quad (2.20)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0. \quad (2.21)$$

The differential equations hold over the domain Ω . We impose the boundary conditions

$$\phi_f \mathbf{v}_r \cdot \nu = \phi_f^{1+\Theta} \tilde{\mathbf{v}}_r \cdot \nu = g_r \quad \text{and} \quad \mathbf{v}_s = \mathbf{g}_s \quad \text{on } \partial\Omega, \quad (2.22)$$

where ν is the outer unit normal vector and g_r and $\mathbf{g}_s$ are given. A compatibility condition is required, and it is

$$\int_{\partial\Omega} (g_r + \mathbf{g}_s \cdot \nu) ds = 0. \quad (2.23)$$

2.2 Transport of mass and energy

Within the flow field, the two components of mass and the thermodynamic energy of the system are transported.

2.2.1 Mass transport

The bulk transport variable is $\bar{c}$, and it is the phase volume average concentration of component 1. It is defined to be

$$\bar{c} = \phi_f c_{1,f} + \phi_1 c_{1,1} + \phi_2 c_{1,2} = \phi_f c_f + \phi_1 c_s. \quad (2.24)$$

With $M_1 = M_{1,f} + M_{1,1}$ being the total mass of component 1, note that within the REV (Figure 1)

$$\bar{c} = \phi_f c_f + \phi_1 c_s = \frac{V_f}{V} \frac{M_{1,f}}{V_f} + \frac{V_1}{V} \frac{M_{1,1}}{V_1} = \frac{M_{1,f} + M_{1,1}}{V} = \frac{M_1}{V} \quad (2.25)$$

is the total mass per unit volume.

Mass conservative transport of component 1 can be expressed in terms of mixture theory by a standard advection-diffusion equation, namely,

$$\bar{c}_t + \nabla \cdot (\bar{c} \mathbf{v} - \phi_f D \nabla c_f) = 0, \quad (2.26)$$

where D is a diffusion coefficient. Note that we have diffusive mixing only in the fluid, since the solid phases are pure. This equation is difficult to approximate, since it involves the concentration in both the solid and fluid phases (especially in the term $\bar{c} \mathbf{v}$).

There are at least two possible approaches to solving (2.26). A standard approach (used e.g., by Katz [16]) is to define the melt term Γ and split the equation into solid and fluid phases. That is, to consider

$$(\phi_f c_f)_t + \nabla \cdot (\phi_f c_f \mathbf{v}_f - \phi_f D \nabla c_f) = \Gamma, \quad (2.27)$$

$$(\phi_1 c_s)_t + \nabla \cdot (\phi_1 c_s \mathbf{v}_s) = -\Gamma. \quad (2.28)$$

We might approximate $\Gamma = 0$ here and expect the flash calculation later to account for the melting term Γ .

However, we propose a different approach to solving (2.26). We define a partition function κ via

$$\phi_f c_f = \kappa(E, \bar{c}) \bar{c}, \quad \kappa := \frac{\phi_f c_f}{\bar{c}}. \quad (2.29)$$

The coefficient κ will be determined by the phase behavior, and so it depends on $\bar{c}$ and the total thermodynamic internal energy per unit volume E to be discussed later. Note that

$$0 \leq \kappa \leq 1, \quad (2.30)$$

since $\phi_f c_f = \kappa \bar{c} = \kappa(\phi_f c_f + \phi_1 c_s)$.

Because

$$\phi_1 c_s = \bar{c} - \phi_f c_f = (1 - \kappa) \bar{c},$$

we can simplify

$$\bar{c} \mathbf{v} = \phi_f c_f \mathbf{v}_f + \phi_1 c_s \mathbf{v}_s = (\kappa \mathbf{v}_f + (1 - \kappa) \mathbf{v}_s) \bar{c} = \mathbf{v}_e^* \bar{c},$$

where

$$\mathbf{v}_e^* = \kappa \mathbf{v}_f + (1 - \kappa) \mathbf{v}_s = \kappa \mathbf{v}_r + \mathbf{v}_s$$

The transport equation (2.26) then becomes

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e^* \bar{c} - \phi_f D \nabla c_f) = 0. \quad (2.31)$$

Although the diffusion term $\phi_f D |\nabla c_f|$ is perhaps small enough to be neglected in this application, we can further simplify it as

$$\phi_f D \nabla c_f = \phi_f D \nabla \left(\frac{\kappa \bar{c}}{\phi_f} \right) = \kappa D \nabla \bar{c} + \bar{c} D (\nabla \kappa - \kappa \nabla \ln \phi_f).$$

Thus transport equation (2.31) becomes

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e \bar{c} - \kappa D \nabla \bar{c}) = 0, \quad (2.32)$$

where

$$\mathbf{v}_e = \mathbf{v}_e^* - D (\nabla \kappa - \kappa \nabla \ln \phi_f) = \kappa \mathbf{v}_r + \mathbf{v}_s - D (\nabla \kappa - \kappa \nabla \ln \phi_f), \quad (2.33)$$

which has the form of an advection-diffusion equation for $\bar{c}$. The phase behavior flash calculation would then give values for c_f and c_s (as well as ϕ_f and ϕ_1).

2.2.2 Energy transport

The phase specific internal energies are denoted e_j for phase $j \in \{1, 2, f\}$. The total internal energy per unit volume is then

$$E = \bar{\rho} e = \phi_f \rho_f e_f + \phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2. \quad (2.34)$$

Similar to (2.26), E satisfies an advection-diffusion equation representing energy conservation, and by mixture theory, it is

$$E_t + \nabla \cdot (\bar{\rho} e \mathbf{v} - \bar{K} \nabla T) = 0, \quad (2.35)$$

where T is the temperature and K is the thermal conductivity, so

$$\bar{K} = \phi_f K_f + \phi_1 K_1 + \phi_2 K_2. \quad (2.36)$$

Define the partition function $\lambda(E, \bar{c})$ by

$$\phi_f \rho_f e_f = \lambda(E, \bar{c}) E, \quad \lambda := \frac{\phi_f \rho_f e_f}{E}. \quad (2.37)$$

The phase behavior will fix the coefficient λ , so it depends on $\bar{c}$ and E . Note that

$$\phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2 = (1 - \lambda) E \quad \text{and} \quad 0 \leq \lambda \leq 1,$$

so the energy conservation law is

$$E_t + \nabla \cdot (\mathbf{v}_e^E E - \bar{K} \nabla T) = 0, \quad (2.38)$$

where effective velocity is

$$\mathbf{v}_e^E = \lambda \mathbf{v}_f + (1 - \lambda) \mathbf{v}_s = \lambda \mathbf{v}_r + \mathbf{v}_s. \quad (2.39)$$

At each time step, the temperature distribution and its gradients will be obtained directly from the phase behavior equations. Thus, the gradient of $T = T(\bar{c}, E)$ can be computed as

$$\nabla T(\bar{c}, E) = \frac{\partial T}{\partial \bar{c}} \nabla \bar{c} + \frac{\partial T}{\partial E} \nabla E. \quad (2.40)$$

2.3 Phase behavior

Given the total mass $\bar{c}$ and internal energy E , a flash calculation is performed to equilibrate the thermodynamic phase behavior of the components. This results in values for c_f , ϕ_f , ϕ_1 , and T . From these, one computes $\phi_s = 1 - \phi_f$ and $\phi_2 = 1 - \phi_1 - \phi_f$. Recall that $c_s = c_{1,1} = \rho_1$ and $c_{2,2} = \rho_2$ are constant.

Olivine and pyroxene are two usual immiscible crystals in mantle melting process, hence a binary eutectic phase diagram can be used to obtain fractions of different components during flash calculations. Figure 2 shows an simple example for binary eutectic phase behavior in partially melting mantle. This binary eutectic phase behavior is well documented and described, and there are established research codes like FREZCHEM [18]. In our application, a simplified phase code is introduced for calculation, which linearizes some of the phase behavior. An important example application, introduced by Marc Hesse [14], is ice shell melting. Ice and hydrohalite are two solid phases in this ice shell melting process, which corresponds to olivine and pyroxene in mantle. The phase diagram generated by our simplified code is illustrated in figure 3 plotted in terms of temperature T and the bulk mass fraction of halite in the system X representing this ice-salt system. By giving the current enthalpy, composition of one component, phase densities, heat capacities and other phase information, we can calculate temperature, phase volume fraction, and corresponding classification of field ions in the phase diagram for any specific binary eutectic system, in our case, the partial melting mantle system.

2.4 The coupled system of governing equations

The complete set of governing equations is summarized here. First, given ϕ_f and $\phi_s = 1 - \phi_f$, we have the Darcy-Stokes flow system (2.18)–(2.21), (2.11) for the scaled variables, which give $\mathbf{v}_r$, $\mathbf{v}_s$, and $\mathbf{v}_f = \mathbf{v}_r + \mathbf{v}_s$ via (2.17). Second, given ϕ_f , $\mathbf{v}_r$, $\mathbf{v}_s$, and T , conservation of mass and energy under transport is determined by (2.32)–(2.33) and (2.38)–(2.39) (and (2.40)), using $\kappa = \kappa(\bar{c}, E)$ defined by (2.29) and using $\lambda = \lambda(\bar{c}, E)$ defined by (2.37). These equations give $\bar{c}$ and E . Finally, the flash calculation takes $\bar{c}$ and E and gives ϕ_f and T , as well as c_f and ϕ_1 .

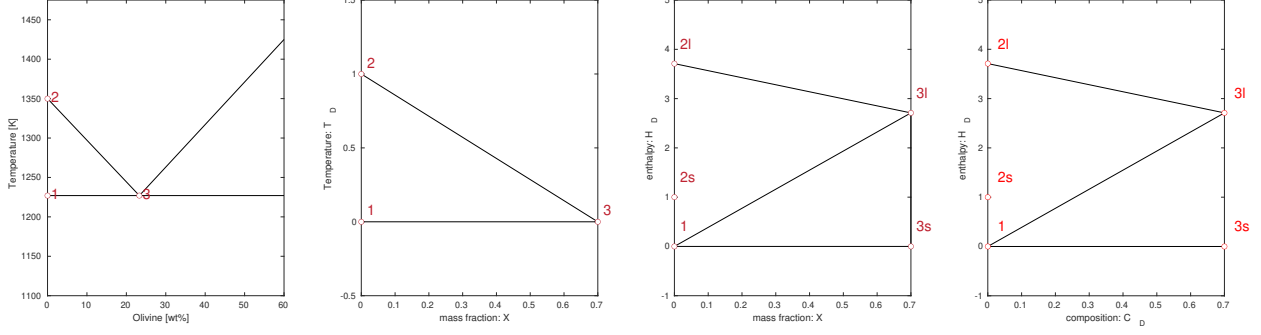


Figure 2: Illustration of binary eutectic phase behavior of partially melting mantle: **a)** Phase diagram, K . **b)** Dimensionless temperature versus mass fraction, T_D . **c)** Dimensionless enthalpy versus mass fraction, H_D . **d)** Dimensionless enthalpy versus composition H_D , ϕ_{bri} .

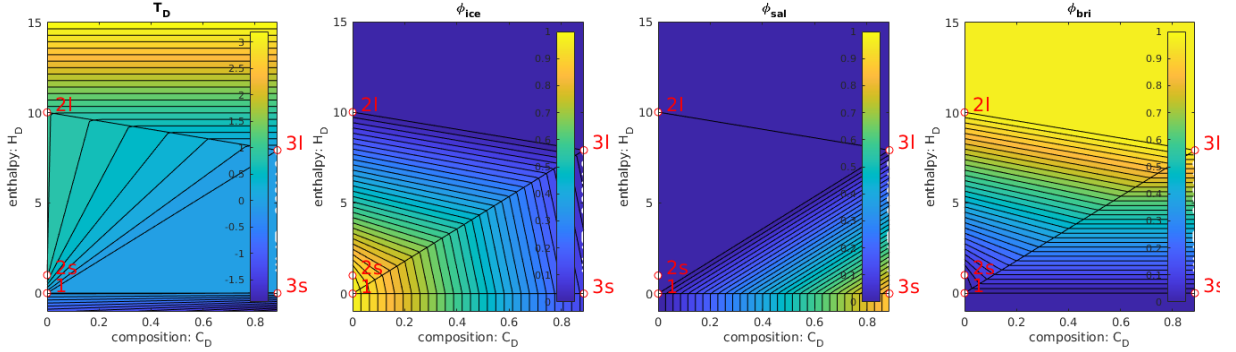


Figure 3: Variation of various phase properties in $\mathcal{CH}$ phase diagram: **a)** Dimensionless temperature, $\mathcal{T}$. **b)** Ice volume fraction, ϕ_{ice} . **c)** Salt volume fraction, ϕ_{sal} . **d)** Brine volume fraction, ϕ_{bri} .

3 Spatial and time discretization

In general, we need to solve the Stokes-Darcy flow equations coupled to a system of two transport equations for mass and energy, and perform a flash calculation. The mixture model used does not require the domain Ω to be split into subdomains representing single and two-phase flows.

A conforming mesh of quadrilaterals will be used in all computations. Discretization on a quadrilateral mesh requires fewer elements than discretization on a triangular mesh. Each quadrilateral element Q should be strictly convex and not degenerate into a triangle, line, or point. For ease of implementation, we will use a logically rectangular mesh and ordering of the elements. An appropriate mesh quality check should verify that these requirements are satisfied before calculations commence. We let $h > 0$ denote the maximal element diameter.

We use mixed finite elements to discretize the flow system and weighted essentially non oscillatory (WENO) methods to solve the transport system. The phase behavior is assumed to be controlled by a eutectic binary phase diagram, and the flash package, i.e., the flash calculation, is an explicit, direct parametrization of the phase diagram (so no auxiliary system of equations needs to be solved). We plan to use third order WENO methods for transport, so the velocities from the flow system should be solved as accurately. However, flow is generally smoother than transport, so we will use only a second order accurate method for the flow velocities.

For ease of presentation of the finite element spaces, we depict in Figure 4 a quadrilateral element Q , which can be viewed as the image of the reference element $\hat{Q} = [0, 1]^2$ under a bilinear

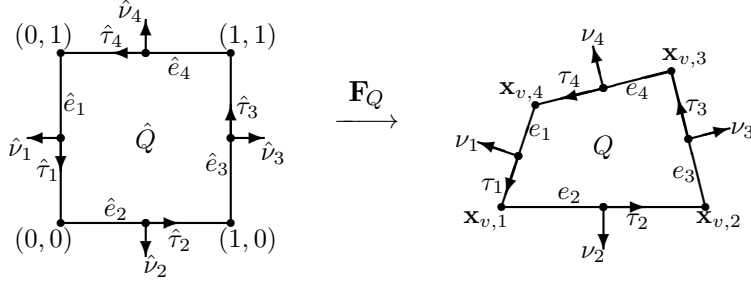


Figure 4: The unit reference element $\hat{Q} = [0, 1]^2$ and quadrilateral Q , with edges $\hat{e}_i$ and e_i , outer unit normals $\hat{\nu}_i$ and ν_i , tangents $\hat{\tau}_i$ and τ_i , and vertices $(-1, -1)$ and $\mathbf{x}_{v,1}$, etc., respectively.

mapping $\mathbf{F}_Q$. We locally index the edges e_i in the counterclockwise direction, and define the outer unit normals as ν_i , the tangents as τ_i , and the vertices as $\mathbf{x}_{v,i}$, for $i = 1, 2, 3, 4$.

3.1 Discretization of the Darcy-Stokes system

Before delving into the details of the discretization, we give a general finite element approximation of the scaled equations (2.18)–(2.21). Let the finite element space for the Darcy part of the flow system be denoted $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$, while $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ denotes the space for the Stokes system. Essential boundary conditions will be imposed using projections into the finite element spaces $\hat{\mathbf{g}}_r \in \mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s \in \mathbb{V}_{S,h}$. The following two compatibility conditions must hold:

$$\int_{\partial\Omega} (\hat{\mathbf{g}}_s - \mathbf{g}_s) \cdot \nu \, ds = 0 \quad \text{and} \quad \int_{\partial\Omega} (\phi_f^{1+\Theta} \hat{\mathbf{g}}_r + \hat{\mathbf{g}}_s) \cdot \nu \, ds = 0. \quad (3.1)$$

We need to conserve the total mass (2.8) of the system. A straightforward finite element approximation of the scaled equations (2.18)–(2.21) does not lead to a mass conservative scheme; however, a conservative scheme can be defined [3]. We propose to use a somewhat simpler formulation than appears in [3].

The approximation of (2.18)–(2.21) is: Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h} + \hat{\mathbf{g}}_r$, $\tilde{q}_{f,h} \in \mathbb{W}_{D,h}$, $\mathbf{v}_{s,h} \in \mathbb{V}_{S,h} + \hat{\mathbf{g}}_s$, and $q_h \in \mathbb{W}_{S,h}$ such that

$$\left(\frac{\mu_f}{k_0} \tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r \right) - (\tilde{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \boldsymbol{\psi}_r)) = 0, \quad (3.2)$$

$$(\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_{r,h}), \omega_f) + \left(\frac{1}{\mu_s(1-\phi_f)} (\tilde{q}_{f,h} - \phi_f^{1/2} q_h), \omega_f \right) = 0, \quad (3.3)$$

$$(q_h, \nabla \cdot \boldsymbol{\psi}_s) - (\hat{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla \boldsymbol{\psi}_s) = ((1-\phi_f) \rho_r \mathbf{g}, \boldsymbol{\psi}_s), \quad (3.4)$$

$$(\nabla \cdot \mathbf{v}_{s,h}, \omega) - \left(\frac{\hat{\phi}_f^{1/2}}{\mu_s(1-\phi_f)} (\tilde{q}_{f,h} - \phi_f^{1/2} q_h), \omega \right) = 0, \quad (3.5)$$

for all test functions $\boldsymbol{\psi}_r \in \mathbb{V}_{D,h}$, $\omega_f \in \mathbb{W}_{D,h}$, $\boldsymbol{\psi}_s \in \mathbb{V}_{S,h}$, and $\omega \in \mathbb{W}_{S,h}$. The straightforward formulation would use ϕ_f in place of $\hat{\phi}_f$ (in three places). On each mesh element Q , the average of ϕ_f on Q is denoted $\phi_{f,Q}$, and this quantity is defined over Ω by $\hat{\phi}_f$, i.e., $\hat{\phi}_f|_Q = \phi_{f,Q}$.

It is important to avoid division by zero in the formulation. Since $\phi_f \in [0, 1]$, but not close to 1, the only terms of concern are the two involving $\hat{\phi}_f^{-1/2}$. On an element Q , $\phi_{f,Q}^{-1/2}$ is constant, and so vanishes if and only if ϕ_f vanishes on Q . In that case, the two terms involving $\phi_{f,Q}^{-1/2}$ should

be interpreted as zero (as can be justified by a scaling argument, since there are more positive powers of ϕ_f than negative powers). Moreover, these terms can be computed using the divergence theorem:

$$\begin{aligned} \int_Q \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \psi_{r,h}) \omega_f dx &= \int_{\partial Q} \phi_{f,Q}^{-1/2} \phi_f^{1+\Theta} \psi_{r,h} \cdot \nu \omega_f ds \\ &\quad - \int_Q \phi_{f,Q}^{-1/2} \phi_f^{1+\Theta} \psi_{r,h} \cdot \nabla \omega_f dx. \end{aligned} \quad (3.6)$$

The last term vanishes when ω_f is constant on Q .

We show the mass conservation assuming that the potentials will be approximated in function spaces containing piecewise discontinuous constants. We define for any mesh element Q the test function ω_Q being 1 on Q and 0 elsewhere. Then the sum of (3.3) with $\omega_f = \phi_{f,Q}^{-1/2} \omega_Q \in \mathbb{W}_{D,h}$ and (3.5) with $\omega = \omega_Q \in \mathbb{W}_{S,h}$ is

$$\int_Q \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) dx + \int_Q \nabla \cdot \mathbf{v}_s dx = 0, \quad (3.7)$$

which is the local mesh element conservation of mass (2.8). We remark that the somewhat more involved formulation in [3] also gives mass conservation of the compaction relation (2.10).

In the next subsections we describe the finite elements that we use. They have piecewise discontinuous constant potential approximations. We also show how to recover the unscaled approximate velocity $\mathbf{v}_{r,h}$ and potential $q_{f,h}$.

3.1.1 Finite elements for the Darcy subsystem

As mentioned above, we use second order accurate elements. In order to maintain sufficient accuracy on the quadrilateral mesh for the Darcy part of the system, we will use a variant of the lowest order reduced $H(\text{div})$ -approximation AC space [2]. The variant uses the direct mixed space $\mathbb{V}_1^0$ given in [6], which is defined directly on each quadrilateral *without* requiring any mapping from a reference element. This space will give a second order accurate approximation of the velocity when paired with a reduced accuracy, piece-wise constant approximation of the pressure (i.e., potential). On a quadrilateral Q , the finite element space is

$$\mathbb{V}_{D,h}(Q) = \mathbb{V}_1^0(Q) = \mathbb{P}_1^2 \oplus \text{curl } \mathbb{S}_1(Q) \quad \text{and} \quad \mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0, \quad (3.8)$$

where the supplemental space $\mathbb{S}_1(Q)$ has dimension 2 (so $\dim \mathbb{V}_1^0(Q) = 8$).

We can construct a basis for this mixed finite element space on the quadrilateral Q . We use the notation of Figure 4. First of all, define the linear polynomials

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{vi}) \cdot \nu_i, \quad i = 1, 2, 3, 4, \quad (3.9)$$

for which $\text{curl } \lambda_i = \tau_i$. Then define

$$R_{13}(\mathbf{x}) = \frac{\lambda_1 - \lambda_3}{\lambda_1 + \lambda_3}, \quad R_{24}(\mathbf{x}) = \frac{\lambda_2 - \lambda_4}{\lambda_2 + \lambda_4},$$

and

$$R_1(\mathbf{x}) = \frac{1}{2}(1 - R_{13}), \quad R_3(\mathbf{x}) = \frac{1}{2}(1 + R_{13}), \quad R_2(\mathbf{x}) = \frac{1}{2}(1 - R_{24}), \quad R_4(\mathbf{x}) = \frac{1}{2}(1 + R_{24}). \quad (3.10)$$

The function R_i is 1 on edge e_i , 0 on the opposite edge, and arbitrary on the other two edges. Now

$$\mathbb{S}_1(Q) = \text{span}\{\lambda_2\lambda_4R_{13}, \lambda_1\lambda_3R_{24}\}, \quad (3.11)$$

and so

$$\mathbb{V}_{D,h}(Q) = \mathbb{P}_1^2 \oplus \text{curl } \mathbb{S}_1(Q) = \mathbb{P}_1^2 + \text{curl span}\{\lambda_2\lambda_4R_1, \lambda_2\lambda_4R_3, \lambda_1\lambda_3R_2, \lambda_1\lambda_3R_4\}. \quad (3.12)$$

We define the 8 basis functions with indexing modulo 4. Half of the basis functions are

$$\mathbf{v}_i^l = \text{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\tau_{i+1}\lambda_{i+3} + \tau_{i+3}\lambda_{i+1})R_i - \lambda_{i+1}\lambda_{i+3} \frac{\tau_i\lambda_{i+2} - \tau_{i+2}\lambda_i}{(\lambda_i + \lambda_{i+2})^3}. \quad (3.13)$$

Then it is easy to verify that

$$\mathbf{v}_i^l \cdot \nu_j|_{e_j} = \text{curl}(\lambda_{i+1}\lambda_{i+3}R_i) \cdot \nu_j|_{e_j} = \nabla(\lambda_{i+1}\lambda_{i+3}R_i) \cdot \tau_j|_{e_j} = \begin{cases} 0, & j \neq i \pmod{4}, \\ \ell_i(\mathbf{x}), & j = i, \end{cases} \quad (3.14)$$

where ℓ_i is a non vanishing linear polynomial on e_i orthogonal to one, i.e., $\int_{e_i} \ell_i ds = 0$.

The remaining 4 basis functions are better defined in a two-stage process. First, let

$$\begin{aligned} \mathbf{v}_i^{c,*} &= \tau_{i+2}\lambda_{i+3} + \tau_{i+3}\lambda_{i+2} - \alpha\mathbf{v}_i^l - \beta\mathbf{v}_{i+1}^l, \\ \alpha &= \frac{(\lambda_{i+3}(\mathbf{x}_{v,i})\tau_{i+2} + \lambda_{i+2}(\mathbf{x}_{v,i})\tau_{i+3} - \lambda_{i+2}(\mathbf{x}_{v,i+3})\tau_{i+3}) \cdot \nu_i}{(\mathbf{v}_i^l(\mathbf{x}_{v,i}) - \mathbf{v}_i^l(\mathbf{x}_{v,i+3})) \cdot \nu_i}, \\ \beta &= \frac{(\lambda_{i+3}(\mathbf{x}_{v,i+1})\tau_{i+2} - \lambda_{i+3}(\mathbf{x}_{v,i})\tau_{i+2} - \lambda_{i+2}(\mathbf{x}_{v,i})\tau_{i+3}) \cdot \nu_{i+1}}{(\mathbf{v}_{i+1}^l(\mathbf{x}_{v,i+1}) - \mathbf{v}_{i+1}^l(\mathbf{x}_{v,i})) \cdot \nu_{i+1}}, \end{aligned} \quad (3.15)$$

for which the normal component on each edge $\mathbf{v}_i^{c,*} \cdot \nu_j|_{e_j}$ is linear. Easily $\mathbf{v}_i^{c,*} \cdot \nu_j|_{e_j} = 0$ for $j = i+2, i+3 \pmod{4}$. In fact, $\mathbf{v}_i^{c,*} \cdot \nu_j|_{e_j}$ is constant for $j = i, i+1 \pmod{4}$. To see this, compute its value at each end of e_j . For e_i , this is

$$\begin{aligned} \mathbf{v}_i^{c,*}(\mathbf{x}_{v,i}) \cdot \nu_i &= (\lambda_{i+3}(\mathbf{x}_{v,i})\tau_{i+2} + \lambda_{i+2}(\mathbf{x}_{v,i})\tau_{i+3}) \cdot \nu_i - \alpha\mathbf{v}_i^l(\mathbf{x}_{v,i}) \cdot \nu_i \\ &= \mathbf{v}_i^{c,*}(\mathbf{x}_{v,i+3}) \cdot \nu_i = \lambda_{i+2}(\mathbf{x}_{v,i+3})\tau_{i+3} \cdot \nu_i - \alpha\mathbf{v}_1^l(\mathbf{x}_{v,i+3}) \cdot \nu_i, \end{aligned}$$

with the equality resulting from the definition of α . For e_{i+1} , we have defined β so that

$$\begin{aligned} \mathbf{v}_i^{c,*}(\mathbf{x}_{v,i+1}) \cdot \nu_{i+1} &= \lambda_{i+3}(\mathbf{x}_{v,i+1})\tau_{i+2} \cdot \nu_{i+1} - \beta\mathbf{v}_2^l(\mathbf{x}_{v,i+1}) \cdot \nu_{i+1} \\ &= \mathbf{v}_i^{c,*}(\mathbf{x}_{v,i}) \cdot \nu_{i+1} = (\lambda_{i+3}(\mathbf{x}_{v,i})\tau_{i+2} + \lambda_{i+2}(\mathbf{x}_{v,i})\tau_{i+3}) \cdot \nu_{i+1} - \beta\mathbf{v}_2^l(\mathbf{x}_{v,i}) \cdot \nu_{i+1}. \end{aligned}$$

Finally, we define

$$\mathbf{v}_i^c = \mathbf{v}_i^{c,*} - \gamma(\mathbf{x} - \mathbf{x}_{v,i+2}), \quad \gamma = \frac{\mathbf{v}_i^{c,*}(\mathbf{x}_{v,i+1}) \cdot \nu_{i+1}}{(\mathbf{x}_{v,i+1} - \mathbf{x}_{v,i+2}) \cdot \nu_{i+1}}, \quad (3.16)$$

so that

$$\mathbf{v}_i^c \cdot \nu_j|_{e_j} = \begin{cases} 0, & j \neq i \pmod{4}, \\ c_i, & j = i, \end{cases} \quad (3.17)$$

for some nonzero constant c_i .

The velocity basis functions can be merged together with continuous normal flux across the element boundaries, so they are $H(\text{div})$ conforming. One omits the basis functions that have nonvanishing normal flux on $\partial\Omega$ to define the global space $\mathbb{V}_{D,h}$. The space $\mathbb{W}_{D,h}$ consists of piecewise constant polynomials. The accuracy of these elements was discussed in [6].

3.1.2 Recovery of the unscaled Darcy variables

The unscaled Darcy (fluid) potential is $q_f = \phi^{-1/2} \tilde{q}$ (see (2.17)). Although probably not needed in any of the other calculations, we can define an approximate Darcy potential. We will assume that $q_{f,h} \in \mathbb{W}_{D,h}$, so on Q we define

$$q_{f,h}|_Q = \begin{cases} \phi_{f,Q}^{-1/2} \tilde{q}_{f,h}|_Q & \text{if } \phi_{f,Q} \neq 0, \\ 0 & \text{if } \phi_{f,Q} = 0, \end{cases} \quad (3.18)$$

where the value of $q_{f,h}|_Q$ is not important when $\phi_{f,Q} = 0$, since in that case there is no fluid phase.

The relative Darcy velocity is given by $\mathbf{v}_r = \phi_f^\Theta \tilde{\mathbf{v}}_r$ (see (2.17)). It would be convenient to assume that $\mathbf{v}_{r,h} \in \mathbb{V}_{D,h}$. In that case, we need merely define the normal trace of $\mathbf{v}_{r,h}$. To maintain local mass balance (3.7), we need to enforce that on each edge $e \subset \partial Q$

$$\int_e \phi_f^{1+\Theta} \tilde{\mathbf{v}}_{r,h} \cdot \nu \, ds = \int_e \phi_f \mathbf{v}_{r,h} \cdot \nu \, ds. \quad (3.19)$$

This is achieved by using a weighted L^2 projection on the edge e , denoted as $P_1^{\phi_f} : L^2(e) \rightarrow \mathbb{P}_1(e)$ and defined by the requirement that for $f \in L^2(e)$, $P_1^{\phi_f} f \in \mathbb{P}_1(e)$ and

$$\int_e \phi_f (f - P_1^{\phi_f} f) \ell \, ds = 0 \quad \forall \ell \in \mathbb{P}_1(e). \quad (3.20)$$

Then

$$\mathbf{v}_{r,h} \cdot \nu = P_1^{\phi_f} (\phi_f^\Theta \tilde{\mathbf{v}}_r \cdot \nu), \quad (3.21)$$

defines $\mathbf{v}_{r,h} \in \mathbb{V}_{D,h}$ on all of the element Q , although later we will only need the normal traces in the calculation of transport.

3.1.3 Finite elements for the Stokes subsystem

For the Stokes part, we need to specify $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$. The velocity should be approximated with first order polynomials in order to obtain second order accuracy. This requirement would suggest that we use the Bernardi-Raugel elements [12]. While this is the natural choice in terms of accuracy, it is not clear that it is preferred in implementation to using simple tensor product spaces. We propose to show that it is a good choice when coupled to the Darcy subsystem. To achieve consistency with the Darcy problem, the same direct serendipity space will be used to define basis function on the same quadrilateral mesh.

Bernardi-Raugel elements can be viewed as the $Q_1 - P_0$ element enriched with four second order supplementary functions. Conventionally, the Bernardi-Raugel element is defined on the reference square and indirectly on quadrilaterals by a bilinear mapping. We will modify the definition so the basis functions are defined directly on the quadrilateral using a direct serendipity space [6]. This is

$$\mathbb{DS}_1(Q) = \mathbb{P}_1 \oplus \mathbb{S}_1(Q), \quad (3.22)$$

where $\mathbb{S}_1(Q)$ was defined previously in (3.11). On a quadrilateral Q , the modified Bernardi-Raugel finite element is

$$\mathbb{V}_{S,h}(Q) = \mathbb{DS}_1^2(Q) \oplus \text{span}\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4\} \quad \text{and} \quad \mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0, \quad (3.23)$$

where the four supplemental functions are defined modulo 4 by

$$\mathbf{b}_i(\mathbf{x}) = \frac{\varphi_i(\mathbf{x})}{\varphi_i(\mathbf{x}_i)} \nu_i, \quad \mathbf{x}_i = \frac{1}{2}(\mathbf{x}_{v,i-1} + \mathbf{x}_{v,i}), \quad \varphi_i = \lambda_{i+1} \lambda_{i+3} R_i, \quad i = 1, 2, 3, 4. \quad (3.24)$$

We remark that the scalar functions φ_i lie in $\mathbb{DS}_2(Q)$.

The remaining eight nodal basis functions are defined in $\mathbb{DS}_1^2(Q)$. They are clear once $\mathbb{DS}_1(Q)$ is defined. Set the auxiliary rational function

$$r(\mathbf{x}) = \frac{\lambda_3(\mathbf{x})\lambda_4(\mathbf{x})}{\lambda_3(\mathbf{x}_{v,1})\lambda_4(\mathbf{x}_{v,1})} - \frac{\lambda_1(\mathbf{x})\lambda_4(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,2})\lambda_4(\mathbf{x}_{v,2})} + \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})} - \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,4})\lambda_3(\mathbf{x}_{v,4})} \quad (3.25)$$

and then define

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=1}^4 r(\mathbf{x}_{e,i}) \frac{\varphi_i(\mathbf{x})}{\varphi_i(\mathbf{x}_i)}. \quad (3.26)$$

Let $\nu_{d,i}$ be a unit normal to the first diagonal joining $\mathbf{x}_{v,i}$ with $\mathbf{x}_{v,i+2}$, and let the corresponding linear polynomials be

$$\lambda_{d,1} = -(\mathbf{x} - \mathbf{x}_{v,1}) \cdot \nu_{d,1} \quad \text{and} \quad \lambda_{d,2} = -(\mathbf{x} - \mathbf{x}_{v,2}) \cdot \nu_{d,2}. \quad (3.27)$$

The four nodal basis functions of $\mathbb{DS}_1(Q)$, one for each vertex, are

$$\psi_{v,1}(\mathbf{x}) = \frac{\lambda_{d,2}(\mathbf{x}) - \frac{1}{2}\lambda_{d,2}(\mathbf{x}_{v,3})(1 + R(\mathbf{x}))}{\lambda_{d,2}(\mathbf{x}_{v,1}) - \lambda_{d,2}(\mathbf{x}_{v,3})}, \quad (3.28)$$

$$\psi_{v,2}(\mathbf{x}) = \frac{\lambda_{d,1}(\mathbf{x}) - \frac{1}{2}\lambda_{d,1}(\mathbf{x}_{v,4})(1 + R(\mathbf{x}))}{\lambda_{d,1}(\mathbf{x}_{v,2}) - \lambda_{d,1}(\mathbf{x}_{v,4})}, \quad (3.29)$$

$$\psi_{v,3}(\mathbf{x}) = \frac{\lambda_{d,2}(\mathbf{x}) - \frac{1}{2}\lambda_{d,2}(\mathbf{x}_{v,1})(1 + R(\mathbf{x}))}{\lambda_{d,2}(\mathbf{x}_{v,3}) - \lambda_{d,2}(\mathbf{x}_{v,1})}, \quad (3.30)$$

$$\psi_{v,4}(\mathbf{x}) = \frac{\lambda_{d,1}(\mathbf{x}) - \frac{1}{2}\lambda_{d,1}(\mathbf{x}_{v,2})(1 + R(\mathbf{x}))}{\lambda_{d,1}(\mathbf{x}_{v,4}) - \lambda_{d,1}(\mathbf{x}_{v,2})}. \quad (3.31)$$

These merge together to give an $H^1(\Omega)$ conforming space over Ω , and we omit the basis functions associated to $\partial\Omega$. Their tensor product, together with the basis functions of (3.24) (again omitting the basis functions associated to $\partial\Omega$) give our modified Bernardi-Raugel finite element space.

The accuracy of the classical Bernardi-Raugel element has been widely discussed [9]. However, the modified element using the direct serendipity space is first defined here. We propose further numerical experiments and thorough research on these new elements, as well as extension to three dimensional cuboidal hexahedra.

3.2 Discretization of the transport problem

The mass and energy transport problem (2.32) and (2.38) is formulated as system of two parabolic advection-diffusion equations in terms of the variable $U = (\bar{c}, E)$ row-wise as

$$U_t + \operatorname{div}[F(U) - A(U) \operatorname{grad} U] = 0, \quad (3.32)$$

where the flux and diffusion coefficient matrices are

$$F(U) = F\begin{pmatrix} \bar{c} \\ E \end{pmatrix} = \begin{pmatrix} \mathbf{v}_e^T \bar{c} \\ (\mathbf{v}_e^E)^T E \end{pmatrix} \quad \text{and} \quad A(U) = \begin{bmatrix} \kappa D & 0 \\ \frac{\partial T}{\partial \bar{c}} & \frac{\partial T}{\partial E} \end{bmatrix}, \quad (3.33)$$

using (2.40). In terms of the average of U over the element Q ,

$$U_Q(t) = \begin{pmatrix} \bar{c}_Q(t) \\ E_Q(t) \end{pmatrix}, \quad \bar{c}_Q(t) = \frac{1}{|Q|} \int_Q \bar{c}(x, t) dx, \quad \text{and} \quad E_Q(t) = \frac{1}{|Q|} \int_Q \bar{E}(x, t) dx, \quad (3.34)$$

the basic finite volume scheme is formulated by also averaging (3.32), so

$$U_{Q,t}(t) = -\frac{1}{|Q|} \int_Q \operatorname{div} [F(U) - A(U) \operatorname{grad} U] dx = -\frac{1}{|Q|} \int_{\partial Q} [F(U)\nu - A(U)\nabla U \cdot \nu] ds, \quad (3.35)$$

after applying the divergence theorem. These boundary integrals will be approximated edge by edge using two point Gauss quadrature.

The value of U on the edge is reconstructed from the element averages through (implicit) finite difference WENO-AO methods [4]. We will use tensor product WENO-AO(3,2) reconstructions [5], which are third order accurate for smooth solutions and drop to second order near discontinuities. The diffusive flux requires reconstruction of the normal derivatives of the solution, so we will use tensor product WENO-AO(4,3) reconstructions [5] in this case. The derivatives of U are then also third order accurate for smooth solutions and drop to second order near discontinuities.

We employ the Local Lax-Friedrichs numerical flux to stabilize the hyperbolic part of the operator. That is, we replace the normal flux

$$F(U)\nu = \begin{pmatrix} \mathbf{v}_e \cdot \nu \bar{c} \\ \mathbf{v}_e^E \cdot \nu E \end{pmatrix}$$

by $\hat{F}(U^-, U^+)$, where U^- and U^+ are one-sided limits of the approximate solution. These limits naturally arise from using WENO-AO(3,2) reconstructions. The Jacobian of the normal flux is

$$\frac{\partial F(U)\nu}{\partial U} = \begin{bmatrix} \frac{\partial(\mathbf{v}_e \cdot \nu \bar{c})}{\partial \bar{c}} & \frac{\partial(\mathbf{v}_e \cdot \nu \bar{c})}{\partial E} \\ \frac{\partial(\mathbf{v}_e^E \cdot \nu E)}{\partial \bar{c}} & \frac{\partial(\mathbf{v}_e^E \cdot \nu E)}{\partial E} \end{bmatrix} \approx \begin{bmatrix} \mathbf{v}_e \cdot \nu & 0 \\ 0 & \mathbf{v}_e^E \cdot \nu \end{bmatrix}. \quad (3.36)$$

The Lax-Friedrichs numerical flux is then

$$\hat{F}(U^-, U^+) = \frac{1}{2} [F(U^-)\nu + F(U^+)\nu - \alpha_{\text{LF}}(U^+ - U^-)], \quad (3.37)$$

where

$$\alpha_{\text{LF}} = \begin{bmatrix} \|\mathbf{v}_e \cdot \nu\|_{L^\infty} & 0 \\ 0 & \|\mathbf{v}_e^E \cdot \nu\|_{L^\infty} \end{bmatrix}. \quad (3.38)$$

We plan to use a local version of α_{LF} .

We discretize time into levels $t^0 < t^1 < t^2 < \dots$ and set $\Delta t^{n+1} = t^{n+1} - t^n$. The time derivative will be integrated over the interval $[t^n, t^{n+1}]$ using a third order Runge-Kutta time stepping method. We choose to use the third order implicit Radau-IIA method, which has the Butcher Tableau shown in Table 1. The implementation of an implicit Runge-Kutta method requires solution of a nonlinear system of equations, which we solve by Newton iteration at each time step.

3.3 Solution of the coupled Darcy-Stokes, transport, and phase behavior system

Let U^n be the solution to the transport system at time t^n . Let $V^n = (\mathbf{v}_{h,r}^n, \mathbf{v}_{h,s}^n, q_{h,f}^n, q_h^n)$ denote the solution to the Darcy-Stokes system at the same time. Moreover, let $W^n = (\phi_f^n, c_f^n, T^n, \phi_1^n)$ denote the output of the flash calculation at t^n . We formulate two overall solution methods. Algorithm 1 couples the equations sequentially. Algorithm 2 couples the equations within the Newton iteration.

Table 1: The Butcher Tableau of the Radau-IIA Runge-Kutta method.

1/3	5/12	-1/12
1	3/4	1/4
	3/4	1/4

Algorithm 1: Sequential solution

Let U^0 be given as initial data.

Initialize:

1. Given U^0 , compute W^0 by the flash calculation.
2. Given ϕ_f^0 in W^0 , solve the Darcy-Stokes system (3.2)–(3.5) for V^0 .

for Time level $n = 0, 1, \dots, n_{\max} - 1$ **do**

1. Given W^n and V^n , define

$$\kappa^n = \frac{\phi_f^n c_f^n}{\bar{c}^n} \quad \text{and} \quad \lambda^n = \frac{\phi_f^n \rho_f e_f^n}{E^n},$$

and then solve the transport system (3.35) for U^{n+1} .

2. Given U^{n+1} , compute W^{n+1} by the flash calculation.
3. Given ϕ_f^{n+1} in W^{n+1} , solve the Darcy-Stokes system (3.2)–(3.5) for V^{n+1} .

end

Both algorithms are initialized from the initial condition on the transport variables U^0 specifying the total initial mass and energy. The flash calculation then gives W^0 . Once we know W^0 , we have ϕ_f^0 and concentration that can compute the initial flow V^0 .

Both algorithms follow the pattern of computing a new U from old V and W , and a new W from the new U . Finally, a new U is computed from ϕ_f in the new W .

For Algorithm 1, U^{n+1} is computed using fixed in time values for the effective velocities $\mathbf{v}_e$ and $\mathbf{v}_e^E$, pre-calculated from the Darcy-Stokes part, and the partition functions κ and λ . These quantities are only updated before commencing the next time step.

Algorithm 2 will use the current information within a monolithic quasi-Newton solve. It still reduces to solving the three subproblems. We simply omit the Newton derivative couplings between the subproblems. However, we will explore improving the partition function approach by linearizing them, i.e., for κ ,

$$\kappa = \kappa^{n,\ell} + d\kappa^{n,\ell}(U^{n,\ell+1} - U^{n,\ell}),$$

and similarly for λ . This will require computing derivatives of κ and λ in the flash package.

Algorithm 2: Iteratively coupled solution

Let U^0 be given as initial data.

Initialize W^0 and V^0 as in Algorithm 1.

for Time level $n = 0, 1, \dots, n_{\max} - 1$ **do**

Set $U^{n,0} = U^n$, $V^{n,0} = V^n$, and $W^{n,0} = W^n$.

for Newton iteration $\ell = 0, 1, \dots$ until convergence (or $\ell_{\max} - 1$) **do**

1. Given $W^{n,\ell}$ and $V^{n,\ell}$, define

$$\kappa^{n,\ell} = \frac{\phi_f^{n,\ell} c_f^{n,\ell}}{\bar{c}^{n,\ell}} \quad \text{and} \quad \lambda^{n,\ell} = \frac{\phi_f^{n,\ell} \rho_f e_f^{n,\ell}}{E^{n,\ell}},$$

and then solve the transport system (3.35) for $U^{n,\ell+1}$. Evaluate U and W at their proper times within the equation (i.e., at time t^n , or at time t^{n+1} using the (n, ℓ) values, or by interpolating these two values to an intermediate time).

2. Given $U^{n,\ell+1}$, compute $W^{n,\ell+1}$ by the flash calculation.

3. Given $\phi_f^{n,\ell+1}$ in $W^{n,\ell+1}$, solve the Darcy-Stokes system (3.2)–(3.5) for $V^{n,\ell+1}$.

end

Set $U^{n+1} = U^{n,\ell+1}$, $V^{n+1} = V^{n,\ell+1}$, and $W^{n+1} = W^{n,\ell+1}$.

end

4 Application to Mantle Dynamics

An important application of developing new advanced algorithm will be new numerical simulation of partial melting beneath mid-ocean ridges. Magmatism along the global network of mid-ocean ridges continuously resurfaces two thirds of the Earth and transfers a significant amount of mass and energy from the mantle to the crust and the oceans [22]. The erupted mid-ocean ridge basalts therefore provide the most continuous sampling of the Earth's upper mantle. Their geochemistry reflects both the chemical heterogeneity of the upper mantle [1, 13, 17] and the dynamic processes of partial melting and melt segregation [20]. The development of new algorithm will meet the challenge of separating contribution of magmatic process from initial chemical heterogeneity of the upper mantle source of these melts, which will further the understand of mid-ocean ridge magmatism. The generated numerical result will be used to compare with current growing data of geochemical observations. The high resolution transport scheme proposed by us will provide a detailed simulation result of melt transport and segregation.

Geochemical evidence for the melting of a heterogeneous mantle is best preserved at mid-ocean ridges that spread very slowly, such as Gakkel ridge or the Southwest Indian ridge [21, 23]. The melting rate of the upper mantle is proportional to the rate of decompression due to mantle upwelling, which is half of the spreading rate. The melt productivity is therefore low due to the slow spreading, while partial melting of fertile heterogeneities becomes the more dominant impact among overall melt composition than melt flux [22]. Meanwhile, geochemical signatures are transported to the surface by porosity waves induced because of this slow spreading environment. Thus the spreading rate is an important parameter need to be carefully monitored in the process of preservation of geochemical signatures from fertile heterogeneities. Advanced numerical simulation will allow us to test the hypothesis that increased spreading rates decrease the preservation of

geochemical signals in melt segregation. This will lead to a better understanding of the dynamics of mid-ocean ridges and hotpots by resolving the small scale dynamics of these processes in solitary waves. Our algorithms will allow the simulation and evolution of melt bands and regions that may even evolve into solitary waves that migrate upward.

Within the mantle, the porosity vanishes in regions where there is no melt, which leads to highly degenerate equations of two-phase flow. The buoyancy of the melt relative to the solid mantle drives melt segregation either by porous flow at a rate governed by the compaction of the solid matrix or high-porosity channels collect beneath the cold lithospheric lid and is deflected towards the mid-ocean ridges, which leads to the formation of a sloping decompaction channel at the base of the lithosphere. By fully utilizing the WENO-AO method and the new Bernardi-Raugel mixed finite element defined above, we expect to complete high fidelity simulation software that can be used to study the dynamics of the partially molten Earth as it evolves in time. The possible degeneracies in the porosity will be well represented and the survival of distinct geochemical signatures in the decompaction channel will also be investigated.

5 Some preliminary numerical results

Part of the code have been completed, hence, some preliminary numerical results are shown in this section.

The new Bernardi-Raugel element was tested with classical trigonometric functions in the two-dimensional case, where the resulting exact solution is given analytically; that is,

$$\begin{aligned} u &= \cos(x) \sin(y), & v &= -\sin(x) \cos(y), \\ \frac{\partial p}{\partial x} &= \cos(x) \sin(y), & \frac{\partial p}{\partial y} &= \sin(x) \cos(y). \end{aligned}$$

The computed numerical result is compared to this exact solution by calculating the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity. We tested the problem on standard rectangular grids (Figure 5) and randomly distorted two dimensional grids (Figure 6).

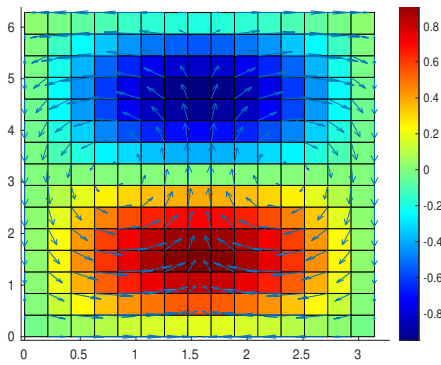


Figure 5: Rectangular Grids. Shown are pressure (color) and velocity (arrows).

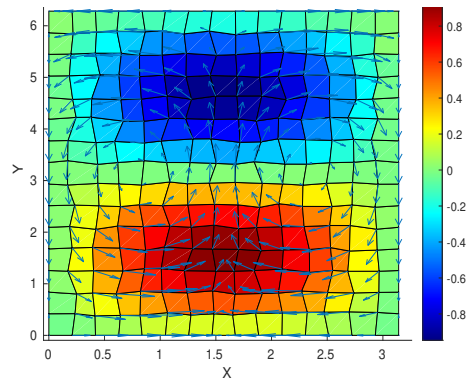


Figure 6: Distorted Grids. Shown are pressure (color) and velocity (arrows).

In Table 2 and Table 3 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. Second order convergence rate is achieved by

velocity and its corresponding derivatives. A first order convergence rate is achieved by pressure. Both meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E-1	1.052E-1	2.513E-2	6.130E-3
$L^2(\text{res}_p)$	1.039E+0	4.463E-1	2.093E-1	1.019E-1
$L^2(\text{res}_{du})$	2.632E+0	5.845E-1	1.374E-1	3.330E-2

Table 2: Rectangular Grids. L^2 -error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E-1	1.074E-2	2.567E-2	6.266E-3
$L^2(\text{res}_p)$	1.070E+0	4.670E-1	2.207E-1	1.078E-1
$L^2(\text{res}_{du})$	2.685E+0	5.981E-1	1.525E-1	4.580E-2

Table 3: Distorted Grids. L^2 -error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

The main code solving 2D problems is written in the PETSc framework, but the implementation of the coupled Darcy-Stokes system is still under development (the above results are from a 2D matlab code). In Figure 7, we display the parallel visualization of the PETSc code by showing a quadrilateral mesh, derived from a random grid distortion of a classic rectangular mesh. It uses the CFD General Notation System (CGNS).

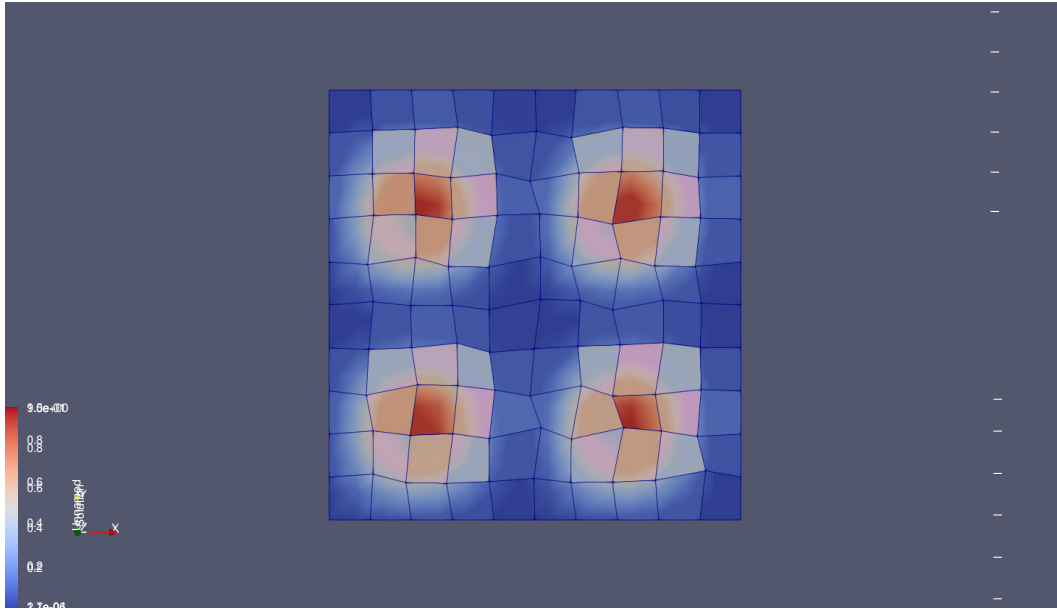


Figure 7: Visualization of a quadrilateral mesh as interfaced with the PETSc-based code.

A simple 1D Burgers equation has been solved with the PETSc-based code, as shown in Figure 8. The implicit WENO3 method was used here. This example serves as a good reference for the more complicated parallel 2D solver, which is under development.

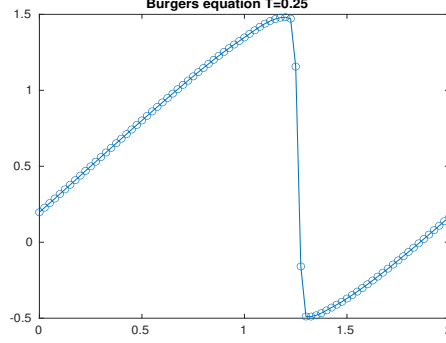


Figure 8: 1D Burgers. Shock formation from a sine wave initial condition.

6 Summary of research topics by areas

We summarize our research contributions in this section, organized by CSEM Research Area.

Area A: Applicable mathematics. We have developed a new and simpler local mass conservative mixed finite element scheme for the Darcy-Stokes system. Moreover, we developed a modified Bernardi-Raugel like element defined using the lowest order direct serendipity space. The stability and convergence of this new element will be studied, and it will be extended to three dimensions. Even though our formulation achieves local mass conservation for the sub problem, we will also analyze it in the context of the complete coupled system of equations, for both Algorithm 1 and Algorithm 2. We will also analyze the error for the flow and transport sub problems separately, as well as propagation of errors within the complete discrete coupled system.

Area B: Scientific computing. A PETSc implementation of the proposed algorithms will be completed. Efficient parallel implementation of implicit WENO-AO and mixed finite element methods will be researched, so the code will handle large-scale, parallel simulations. The suitability of the linearized formulation of κ and λ will be studied from an algorithmic point of view, to see if it indeed both converges and improves the overall performance of the code. A proper stabilizer for the transport problem, will be investigated. We will begin by considering the suitability of the Lax-Friedrichs type numerical flux (3.36), at least in the applications we are targeting. The two different solution algorithms, sequential and iterative coupling, will also be compared and evaluated for their efficiency and accuracy.

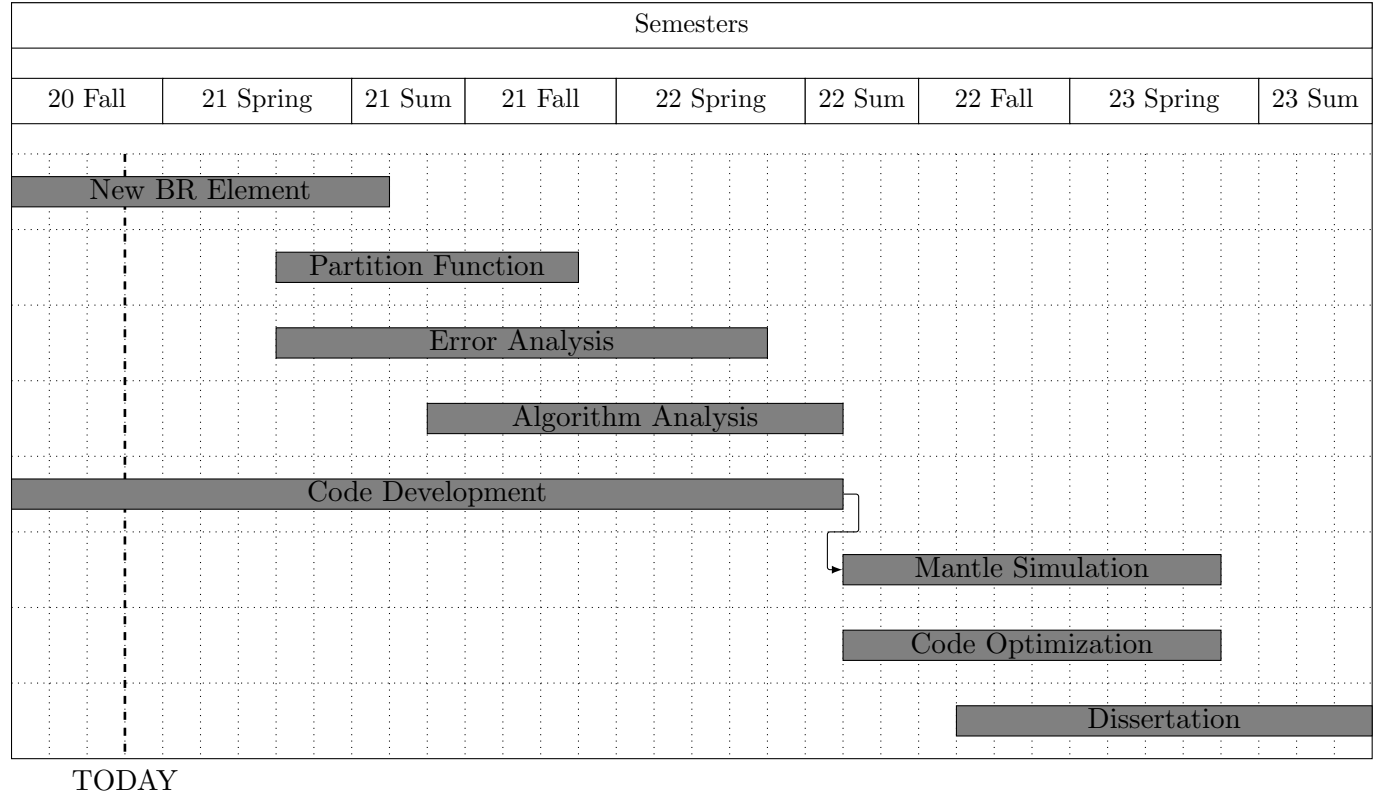
Our code will employ the implicit WENO-AO method, which to our knowledge will be its first use in full-scale mantle simulation. We expect that it will control and reduce numerical oscillation in the solution better than is seen for competing codes.

Area C: Application to Earth and planetary science. The κ and λ formulation will be studied from an applications point of view, to see if it indeed encapsulates enough of the physics that Algorithm 1 could be useful in the applications. Our new partition function approach will be compared with the standard approach proposed by Katz [16]. The new simpler mass conservative mixed method for flow will also be studied from a physical point of view, to see if it adequately preserves the underlying physics. The use of eutectic binary phase behavior, and its corresponding phase package, will be to our knowledge the first time it is used for large-scale mantle simulation. We expect to see a improved approximation of the dynamics of partial melting in simulations of

melt extraction beneath mid-ocean ridges and its geochemical consequences. In particular, we aim to simulate a viscous corner flow with decompression melting and a conductive lid that channels the melt towards the mid-ocean ridge. We will study the effect of chemical heterogeneities with reduced melting point on melt generation and extraction, and their resultant geochemical signatures in the erupted lavas.

7 Timeline of remaining work

The estimated timeline is given below. The main body of code is under development. We are also conducting research on the new Bernardi-Raugel element at the same time. Ideally, we can finish code development before 2022 Fall semester and conduct corresponding mantle simulations for some realistic problems by the end of 2023 Spring semester. Research on propagation of errors, the partition function approach, and analysis of the two algorithms will be conducted along with the code development. The hopeful expectation is that the project and its corresponding dissertation would be complete by the end of the 2023 Summer semester.



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 - Membership in Aerodynamics subsystem